

WASP-12b

R-0382

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_230211 · created 2026-07-18T20:28:04+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459986.8112 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.098 +/- 0.044
Transit depth (Rp/Rs)^2	0.96 +/- 0.85 [%]
Orbital Inclination (inc)	78.0 +/- 2.91
Airmass coefficient 1 (a1)	1.008 +/- 0.044
Airmass coefficient 2 (a2)	-0.008 +/- 0.015
Scatter in the residuals of the lightcurve fit is	4.34 %
Best Comparison Star	#2 - [328, 299]
Optimal Aperture	2.62
Optimal Annulus	13.2
Transit Duration (day)	0.103 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.098 ± 0.044 vs 0.1178 (deviation 0.45σ)
Duration fit vs published	0.103 d vs 0.125 d (deviation 1.38σ)
Mid-time O–C	0.1 min
Depth SNR	2.2
Residual scatter	4.34 %

Light curve

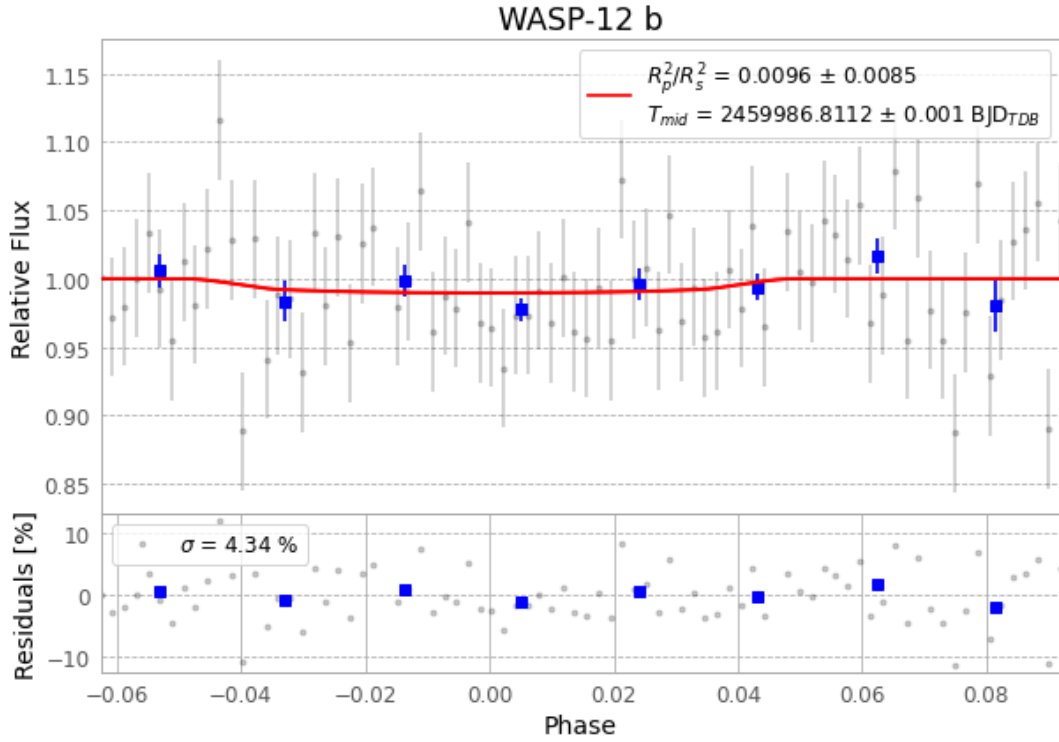


Figure 1 — FinalLightCurve_WASP-12 b_2023-02-10.png (SHA-256 8d9dd1d4d20c54e4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [407, 260], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 499b9e291d9cb089...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8350c6b

Data provenance

85 FITS frames (56 MB), WASP-12230211054215.FITS → WASP-12230211095715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-230211.log (7710adfb94ca...), FinalLightCurve_WASP-12 b_2023-02-10.png (8d9dd1d4d20c...), FinalLightCurve_WASP-12 b_2023-02-10.csv (2d98f476ac5a...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 20:28Z.